

Late-Layer Geometric Resistance Under In-Context Knowledge Conflict

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Status: workshop-note draft · CPU/MPS experiments · 2026-07-06 **Scope:** a narrow, falsification-heavy interpretability finding — see Section 5 before citing.

Abstract

When a document in context contradicts a language model’s parametric knowledge, does that conflict leave a measurable trace in the residual stream? On isolated, single-sentence synthetic contradictions, yes, robustly: across three instruction-tuned models spanning three architecture families (Qwen3-4B-Instruct, Phi-3-mini, Llama-3.2-3B-Instruct) and three relation types (country→capital, company→founder, book→author), a late-layer (~82–83% depth) hidden state for a same-entity contradiction is geometrically further from a clean baseline than switching to a different entity entirely is (9/9 model×relation combinations agree on direction), while a mid-layer (~32–34% depth) check shows no such gap. We then falsify our own first two explanations for *why* the effect varies in size — first that it tracks per-fact parametric memorization strength (rigidity hypothesis, refuted: Spearman rho=0.006, n=45), then that it tracks the model successfully outputting the false claim (active-lie hypothesis, refuted: resisting the contradiction costs *more* geometric disruption than complying with it, n=73 vs 62). Finally — and this is the central limitation of the whole result — the effect is largely an artifact of the single-sentence framing: scaled to 120 facts across 6 relations with realistic multi-sentence documents (the target fact embedded among other true facts, not isolated), the late-layer gap survives for only 1 of 6 relations and *reverses sign* for the other 5. The mid-layer null is separately reconciled with prior work: a *trained* probe recovers a real but modest signal (AUROC 0.642) where our untrained geometric check found none, consistent with — but quantitatively short of — Zhao et al. (2024)’s reported ~90% mid-layer probe accuracy.

1. Setup

Three relations, 15 facts each (45 total), each posed as Document: {fact}. Q: {question} A: with either the true tail (congruent) or another fact’s real tail substituted (contradictory, a plausible hard negative rather than gibberish). Models confirmed to actually follow in-context overrides before use (checked directly, not assumed, and at the *aggregate* level, not on one example — a single Llama-3 test case looked like a premise failure and was not representative, see Section 2.2): GPT-2-base and TinyLlama-1.1B-Chat both failed this premise outright at scale and were excluded; Qwen3-4B-Instruct, Phi-3-mini-4k-instruct, and Llama-3.2-3B-Instruct all pass, with real

(if uneven) override rates on every relation. Llama-3 required a license acceptance and access token, obtained partway through this work.

2. Results

2.1 Real-LLM parametric recall (baseline sanity check)

A linear probe on GPT-2 hidden states, trained on 20 countries’ true/false capital sentences, reaches AUROC 0.980 on 10 held-out countries — confirms parametric facts are linearly decodable and generalize to unseen entities, before testing anything harder.

2.2 Late-layer decoherence under in-context contradiction

Cosine similarity between congruent and contradictory hidden states, same entity, vs. cosine similarity between two different entities’ congruent hidden states (the “does a lie perturb the representation more than an ordinary topic change” test), Qwen3-4B, layers 12/36 and 30/36:

layer	same-entity (congruent vs contradictory)	cross-entity (congruent vs congruent)
12	0.9974	0.9969
30	0.9277	0.9829

Replicated across 3 relations $\times$ 3 seeds (mid-layer gap ~ 0 on all 9 runs, late-layer gap positive on all 9), and on two further, architecturally different models: Phi-3-mini (layers 11/32 and 27/32) and Llama-3.2-3B-Instruct (layers 9/28 and 23/28) — same qualitative pattern on both, 9/9 relation-model combinations now agreeing on direction across all three model families:

model	relation	override %	mid-layer gap	late-layer gap
Qwen3-4B	capital_of	71%	−0.0006	0.0525
Qwen3-4B	founded_by	98%	−0.0035	0.0156
Qwen3-4B	written_by	84%	−0.0048	0.0225
Phi-3-mini	capital_of	22%	−0.0021	0.0214
Phi-3-mini	founded_by	51%	−0.0067	0.0165
Phi-3-mini	written_by	64%	−0.0091	0.0430
Llama-3.2-3B	capital_of	89%	0.0032	0.0612
Llama-3.2-3B	founded_by	16%	−0.0074	0.0816
Llama-3.2-3B	written_by	16%	−0.0156	0.1681

Relation-level magnitude is *not* stable across models: Qwen’s largest effect is **capital_of** (0.053), Phi-3’s largest is **written_by** (0.043), Llama-3’s largest is also **written_by** but at 0.168 — nearly 3 $\times$ Phi-3’s — magnitude depends on the model, not solely the relation, and Llama-3’s effect sizes are uniformly the largest of the three models on this single-sentence framing.

2.3 Rigidity hypothesis — refuted at fact-level granularity

Proposed mechanism: facts held with a “deeper parametric attractor” should decohere more under override. Tested directly: per-fact teacher-forced confidence in the true tail (bare prompt, no

document) against per-fact layer-30 decoherence, $n=45$ facts pooled across all 3 relations. Spearman $\rho = 0.006$ — no relationship. Confidence vs. override-resistance: $\rho = -0.225$, correct direction but below the ~ 0.29 threshold $n=45$ needs for significance. The relation-level magnitude pattern in Section 2.2 is real but not explained by per-fact memorization strength.

2.4 Active-lie hypothesis — refuted with real statistical power

Proposed alternative: decoherence tracks the model successfully *committing* to the false answer, not merely processing the contradiction. Tested on Phi-3 (lower compliance rate gives a naturally large stubborn/compliant split: $n=73$ stubborn, $n=62$ compliant, vs. only $n=5$ stubborn on Qwen):

layer	stubborn (resisted override)	compliant (followed override)	baseline (no conflict)
11	0.0036	0.0036	0.0095
27	0.1018	0.0747	0.0623

Stubborn $>$ compliant $>$ baseline at the late layer — the opposite of what an active-lie mechanism predicts. Resisting an explicit contextual instruction to preserve a parametric prior costs more geometric disruption than complying with it. At the mid layer, both groups score *below* the cross-entity baseline, which is a lexical artifact (swapping the whole subject perturbs shallow-layer tokens more than swapping only the claimed answer for the same subject) rather than evidence about conflict processing.

2.5 Reconciling the mid-layer null with Zhao et al. (2024)

Zhao et al. (arXiv:2410.16090) report $\sim 90\%$ accuracy from a *trained* linear probe detecting conflict-presence at intermediate layers of Llama3-8B — in apparent tension with our untrained mid-layer null (Section 2.2). Replicating their actual method (trained logistic regression, congruent=0/contradictory=1, held out on unseen entities) on our 45-fact Qwen set: AUROC 0.642 at layer 12, 1.000 at layer 30. A trained probe *does* recover a real signal our untrained geometric check missed — the discrepancy is substantially a method difference — but 0.642 falls well short of their $\sim 90\%$, plausibly due to this replication’s much smaller dataset and a single train/held-out split with no cross-validation, not a clean quantitative match.

2.6 Realistic multi-sentence documents — the effect mostly does not survive

Every result above uses an isolated, single-sentence document: **Document: France is a country whose capital is Munich.** Real RAG documents are longer and contain other true content alongside the fact being queried. Scaled to 120 facts across 6 relations (`capital_of`, `founded_by`, `written_by`, `painted_by`, `directed_by`, `composed_by`, 20 facts each) with documents built as an opener + 2 *other true facts from the same relation* + the target sentence + a closer (congruent and contradictory versions share the identical filler facts, differing only in the target sentence’s tail — verified by construction, and a first version of this experiment had a bug letting filler facts differ between the two versions; fixed and rerun before trusting the result below), Qwen3-4B-Instruct:

relation	override %	mid-layer gap	late-layer gap
capital_of	35%	−0.0011	0.0048

relation	override %	mid-layer gap	late-layer gap
founded_by	58%	−0.0032	−0.0105
written_by	88%	−0.0040	−0.0177
painted_by	92%	−0.0039	−0.0179
directed_by	85%	−0.0033	−0.0200
composed_by	95%	−0.0050	−0.0171

The late-layer gap survives for only 1 of 6 relations (`capital_of`, and even there it shrinks roughly $10\times$ from the single-sentence version’s 0.0525 to 0.0048) and *reverses sign* for the other 5. This is the central limitation of the whole result: the effect documented in Sections 2.2–2.4 is largely specific to isolated single-sentence contradictions and does not straightforwardly generalize to documents containing other true content alongside the target fact — the realistic case a deployed RAG verifier would actually need to handle. A plausible mechanism, not yet tested directly: once a document contains multiple true facts alongside one manipulated one, most of the document’s content is identical between the congruent and contradictory versions, which may dominate the late-layer representation and wash out or invert the effect of the one changed sentence relative to switching to a document about a completely different entity.

3. What is and isn’t shown

- On isolated single-sentence contradictions, a real, cross-model (3 architecture families), cross-relation late-layer geometric signal exists and is not explained by per-fact memorization strength or by whether the model ultimately complies.
- **That signal mostly does not survive realistic multi-sentence documents** (Section 2.6) — 1 of 6 relations keeps a (much smaller) positive gap, 5 reverse sign. This is the headline limitation, not a footnote: whatever this signal is, it is substantially a property of isolated single-fact framing, not yet a property of realistic documents.
- The magnitude of the single-sentence effect is model- and relation-dependent by a factor of up to $\sim 3\times$; no mechanism tested so far explains that variation, and it is now moot for 5 of 6 relations once documents are realistic anyway.
- The practical framing that survives on single-sentence documents is **not** “lie detector” — it is closer to a **resistance/friction detector**: it appears to fire more when a model fights its own parametric prior than when it quietly updates to match new context. Whether *any* framing survives on realistic documents is now the open question.
- Not shown: what explains the magnitude variation on single-sentence documents; what explains the sign reversal on realistic documents; whether either transfers past 6 relation templates and 3 model families; a deployed detection threshold (none is proposed here, and Section 2.6 makes one look premature).

4. Honest scope & limitations

- **The realistic-document result (Section 2.6) is the main limitation, not an afterthought.** The signal this note is built around is largely specific to isolated single-sentence contradictions. Anyone citing the Section 2.2–2.4 numbers should cite Section 2.6 in the same breath.
- **Moderate n.** 120 facts / 6 relations for the realistic-document result; 45 facts / 3 relations for the single-sentence cross-model result. Real effects at this scale, not yet large-sample

claims.

- **Two models failed the override premise entirely.** GPT-2-base and TinyLlama-1.1B-Chat do not reliably follow in-context overrides at all and were excluded rather than forced; worth noting as a capability floor for any downstream use of this signal. A single worked example is not sufficient evidence a model passes this premise either way — one Llama-3 example looked like a failure and was not representative of its 45-fact aggregate rate (Section 2.2).
- **No deployed detector.** This note establishes a measurable phenomenon (fragile even on the axis it was found on), not a calibrated, cross-domain threshold ready to gate real generations.
- **Prior art.** Zhao et al. (2024) already show conflict-presence is probeable at intermediate layers; this note’s distinct contributions are the late-layer resistance-vs-compliance split (Section 2.4), the negative result on per-fact rigidity (Section 2.3), and the negative result on realistic-document generalization (Section 2.6) — not the base claim that conflict leaves a residual-stream trace.

5. Reproduce

```
git clone https://github.com/HemanthKiranPolu/nous-decoherence
cd nous-decoherence
pip install -r requirements.txt
python -m nous.verifier_real_llm_test           # 2.1, GPT-2 parametric recall probe
python -m nous.verifier_in_context_conflict     # 2.2, Qwen single-relation decoherence
python -m nous.verifier_multi_relation_robustness # 2.2, 3-relation x 3-seed Qwen sweep
python -m nous.rigidity_hypothesis_test         # 2.3, per-fact confidence vs decoherence
python -m nous.verifier_cross_model_phi3        # 2.2, Phi-3 cross-model replication
python -m nous.verifier_cross_model_llama3      # 2.2, Llama-3 cross-model replication (HF)
python -m nous.verifier_compliance_split_phi3   # 2.4, stubborn vs compliant split
python -m nous.replicate_zhao_conflict_probe    # 2.5, trained-probe reconciliation
python -m nous.verifier_scaled_realistic        # 2.6, 120-fact realistic-document sweep
```

References

- Zhao et al., *Analysing the Residual Stream of Language Models Under Knowledge Conflicts*, arXiv:2410.16090 (2024).
- Li et al., *SHIFT: Gate-Modulated Activation Steering for Knowledge Conflict Mitigation in Retrieval-Augmented Generation*, arXiv:2606.27786 (2026).